

Spacecraft Navigation Project

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1 Introduction

Controlling a real physical system such that it satisfies some specifications is a common problem. In this work, we focus on the *reach-avoid* specification. A concrete example is guiding a spacecraft to some target regions while avoiding hazards.

Such a problem becomes challenging when dealing with uncertainty in the real world, and one might desire quantified statements about satisfying the specifications. One possible approach is to predict the entire uncertainty history with formal guarantees. However, it is not trivial.

In this work, we use stochastic differential equations to model physical systems and explore *general* methods capable of achieving *reach-avoid* goal without exact uncertainty propagation. More importantly, we would like to investigate their limitations and discuss further paths to providing formal guarantees. In case of boring the reader, we quickly summarize the main results:

1. A control policy Π consisting of feed-forward control (from pre-computed nominal trajectory) and feedback regulator (using online optimization) cannot satisfy the specifications due to increasing deviations.
2. Therefore, we present a **control policy** Π^* that generates new nominal trajectory if needed, which is enabled by the efficient Kinodynamic sampling-based planner, SetRRT, we previously developed. Over a limited Monte-Carlo experiment of 100 trials, it achieves 98% of success.
3. The empirical performance of the Π^* can be improved by (i) using accurate dynamics and including uncertainty for the optimization problem of the feedback control, and (ii) informing the planner of the stochastic noise and the feedback control.
4. For the Π^* , we might be able to provide formal guarantees by bounding the deviation error via the knowledge of feedback control and uncertainty-aware planner.

1.1 Related Work

This work is motivated by [1]. The key difference is: **Our approach can satisfy *reach-avoid* specifications, while [1] cannot.** In particular,

- [1] cannot reach *specific target regions* of the asteroid surface because its control policy minimizes the rate of a Lyapunov function V defined on orbit elements deviations. However, this does not guarantee reaching the target orbit elements. This was shown in there experiments.
- [1] discuss the controllability issue because their formulation (i.e., minimizing the rate of V) does not guarantee it to be negative for every time. In contrast, our approach does not suffer from this issue.

- [1] cannot *avoid unsafe regions* because their formulation does not include state constraints.

A non-exhaustive list of related work is provided, along with discussions of the differences from our work.

- A more recent work [2] is proposed to probabilistically satisfy the *reach-avoid* specifications through chance constraints. Although showing good empirical performance, this approach may not yield good results for other stochastic nonlinear systems because it represents uncertainty with mean and covariance only.
- FaSTrack [3] (Fast and Safe tracking) also uses a combination of planning and tracking for *reach-avoid* specifications. However, they model the system as ordinary differential equations with *bounded* disturbances.

2 Problem Formulation

Given dynamical systems modeled as continuous-time nonlinear stochastic differential equations

$$dx = f(x, u, t)dt + \sigma(x, u, t)dw, \quad (1)$$

design a policy $\Pi : X \times T \rightarrow U$ such that the system can satisfy the specifications with high probability. Here, we focus on the *reach-avoid* specifications: eventually reaching the goals while avoiding unsafe regions. In addition, we seek for policies that could be applied during run-time (fast computation).

2.1 Solar-sail Landing Problem

Consider a solar-sail spacecraft that aims to safely land on the surface of an asteroid. Such spacecraft is able to alter the external acceleration induced by the solar radiation pressure (SRP) by changing its attitude. In this work, the dynamical is modeled as the Augmented Normalized Hill Three Body Problem (ANH3BP):

$$\ddot{x} = 2\dot{y} - \frac{x}{r^3} + 3x + \frac{a_x}{a_{\text{norm}}} \quad (2)$$

$$\ddot{y} = -2\dot{x} - \frac{y}{r^3} + \frac{a_y}{a_{\text{norm}}} \quad (3)$$

$$\ddot{z} = -z - \frac{z}{r^3} + \frac{a_z}{a_{\text{norm}}}, \quad (4)$$

where x, y, z are the normalized positions of the spacecraft with respect to the center of the asteroid, $r = \sqrt{x^2 + y^2 + z^2}$, a_x, a_y, a_z are the accelerations from solar radiation pressure, and a_{norm} is the value for acceleration normalization.

Let $\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T$ be the state vector. Let $\mathbf{u} = [u_1, u_2]^T$ be the control vector, where u_1 is the pitch angle and u_2 is the clock angle of the spacecraft attitude. From the solar radiation pressure model

$$a_x = g_0 \cos^2 u_1 (C_1 \cos^2 u_1 + C_2 \cos u_1 + C_3) \quad (5)$$

$$a_y = g_0 \cos^2 u_1 [-(C_1 \cos u_1 + C_2) \sin u_1 \sin u_2] \quad (6)$$

$$a_z = g_0 \cos^2 u_1 [-(C_1 \cos u_1 + C_2) \sin u_1 \cos u_2], \quad (7)$$

where g_0 is the characteristic SRP acceleration, and C_1, C_2, C_3 are the SRP coefficients capturing the reflection.

To land on an asteroid *safely*, the set of final states is $\{\mathbf{x} \in \mathbb{R}^6 : \|x, y, z\|_2 \leq r_{\text{asteroid}} \wedge \|\dot{x}, \dot{y}, \dot{z}\|_2 \leq v_{\text{tol}}\}$, where r_{asteroid} is the normalized asteroid radius, and v_{tol} is the normalized speed tolerance. To incorporate the process noise, the above dynamics is expressed by the SDE in Eq.1. We use Ito's Lemma to simulate the dynamics forward. For the following context, we will abuse the notation, denoting the state vector x , and the control vector u .

3 Feedback Control

In a previous work, we showed that if the process noise w is not present, then we can use sampling-based methods to solve this problem. In this case, the sampling-based methods generate policies that only depend on time, which can be seen as open-loop or feed-forward controllers. However, it is not trivial to explicitly consider w in these methods; it is an active research area.

A common approach to deal with uncertainty w is through feedback controller, such as Linear Quadratic Regulator (LQR), Neighboring Optimal Control (NOC), or their variants. Define $\delta x(t) = x(t) - x^*(t)$ as the deviation between the actual state and the nominal state, then the goal of these feedback controllers is to drive $\delta x(t)$ to zero. Note that a first-order linear approximation of the $\delta x(t)$ dynamics is obtained through the Jacobians:

$$\delta \dot{x}(t) = A(t)\delta x(t) + B(t)\delta u(t), \quad A(t) = \frac{\partial f}{\partial x}, \quad B(t) = \frac{\partial f}{\partial u}. \quad (8)$$

In summary, the policy consists of the nominal feed-forward state and control trajectories $x^*(t), u^*(t)$ and the state feedback controller $\Delta u(t) = K(t, x, u, x^*, u^*)$; the applied control at each time t is $u(t) = u^*(t) + \delta u(t)$. As a motivating example, Fig. 1 illustrates the influence of w for the discussed Solar-sail landing system. The actual trajectory deviates from the nominal trajectory and fails to land on the asteroid. Note that such deviation could also bring the system to unsafe regions.

3.1 Time-invariant Infinite Horizon LQR

We investigate the most naive implementation using discrete-time LQR over infinite horizon. In this formulation, we find optimal δu_k to minimize the cost $J = \sum_{k=0}^{\infty} \delta x_k^T Q \delta x_k + \delta u_k^T R \delta u_k$ subject to the time-invariant discrete-time linear dynamics $x(k+1) = A\delta x_k + B\delta u_k$. The detail implementation is provided below.

At each time step k , we first obtain the discrete-time linear dynamics $\delta x(k+1) = A\delta x_k + B\delta u_k$ by converting the continuous-time linear dynamics

$$\begin{bmatrix} A & B \\ 0 & I \end{bmatrix} = \exp(\Delta t \begin{bmatrix} A(t) & B(t) \\ 0 & 0 \end{bmatrix}). \quad (9)$$

Then, we solve for P_{ss} that should satisfy the discrete time algebraic Riccati equation (DARE)

$$P_{ss} = A^T P_{ss} A - A^T P_{ss} B (R + B^T P_{ss} B)^{-1} B^T P_{ss} A + Q, \quad (10)$$

Lastly, if P_{ss} is solved, we compute the feedback gain

$$K = (R + B^T P_{ss} B)^{-1} B^T P_{ss} A, \quad (11)$$

and the feedback control at this time step is $\delta u_k = -K\delta x_k$. Otherwise δu_k is zero.

Note that we expect this approach to perform poorly because it assumes that the dynamics of δx is time-invariant.

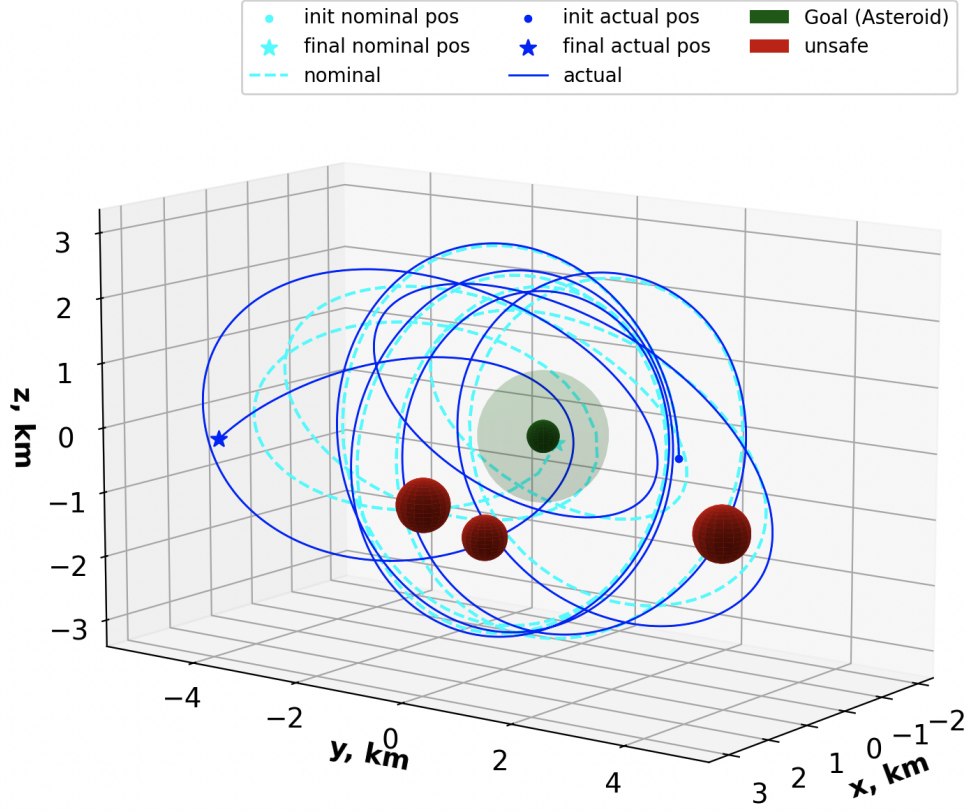


Figure 1: The actual trajectory (blue solid line) deviates from the nominal one (cyan dashed line) due to process noise w .

3.2 Time-varying Finite Horizon LQR

We investigate the discrete-time time-varying LQR over finite horizon. The cost is

$$J = \sum_{k=0}^{N-1} \left(\delta x_k^T Q \delta x_k + \delta u_k^T R \delta u_k \right) + \delta x(N)^T Q_f \delta x(N). \quad (12)$$

To solve for the optimal $\delta u_0, \delta u_1, \dots, \delta u_{N-1}$ that minimizes J above, we use dynamic programming (DP) principle

$$V_k(\delta x_k) = \delta x_k^T Q \delta x_k + \min_{\delta u_k} \left(\delta u_k^T R \delta u_k + V_{k+1}(\mathbf{A}_k \delta x_k + \mathbf{B}_k \delta u_k) \right), \quad (13)$$

where $V_k(\delta x_k) := \min_{\delta u_k, \dots, \delta u_{N-1}} \sum_{k=0}^{N-1} \left(\delta x_k^T Q \delta x_k + \delta u_k^T R \delta u_k \right) + \delta x(N)^T Q_f \delta x(N)$ subject to $\delta x_{k+1} = \mathbf{A}_k \delta x_k + \mathbf{B}_k \delta u_k$, and boundary condition $V_N(\delta x_N) = \delta x_N^T Q_f \delta x_N$. If Q_f is PSD, then $V_k(\delta x_k) = \delta x_k^T P_k \delta x_k$ has a quadratic form, and P_k satisfies

$$P_k = Q + \mathbf{A}_k^T P_{k+1} \mathbf{A}_k - \mathbf{A}_k^T P_{k+1} \mathbf{B}_k (R + \mathbf{B}_k^T P_{k+1} \mathbf{B}_k)^{-1} \mathbf{B}_k^T P_{k+1} \mathbf{A}_k, \text{ subject to } P_N = Q_f. \quad (14)$$

The corresponding control gain

$$K_k = (R + \mathbf{B}_k^T P_{k+1} \mathbf{B}_k)^{-1} \mathbf{B}_k^T P_{k+1}, \quad (15)$$

and the feedback control $\delta u_k = -K_k \delta x_k$ is computed and used for every time step k .

Note that we do not directly use the time-varying control gains $\{K_0, K_1, \dots, K_N\}$ computed at the initial time step ($k=0$) for the future time steps ($k>0$). This reason is simple. Suppose, at time step $k=0$, we use $\delta u_0 = -K_0 \delta x_0$, then the real deviation at the next time step $\delta x_1 \neq \mathbf{A}_0 \delta x_0 + \mathbf{B}_0 \delta u_0$ because

(i) the dynamics of the state deviation is nonlinear and (ii) the existence of process noise. Hence, we have to repeatedly solve the time-varying LQR problem at every time step, where the first control is used. This is the concept of model predictive control (MPC).

We also note that the continuous-time formulation suffers from numerical issues: the values in the matrix $P(t)$ explodes while integrating backward in time

$$\dot{P}(t) = -P(t)A(t) - A^T(t)P(t) + P(t)B(t)R^{-1}B^T(t)P(t) - Q, \text{ subject to } P(t_f) = Q_f, \quad (16)$$

which is likely due to the naive integration scheme $P(t - \Delta t) = P(t) - \dot{P}(t)\Delta t$, where Δt is the integration time step.

3.3 Neighboring Optimal Control

We investigate the neighboring optimal control $\min_{\delta u(t)} \int_{t_0}^{t_f} \delta u^T(t) \delta u(t) dt$ subject to given initial deviation $\delta x(t_0)$ and final boundary condition $\delta x(t_f) = 0$. The idea behind this particular optimization problem is to achieve zero state deviation at the final time (to land on the asteroid) while minimizing the control deviations. If we define the Hamiltonian according to the first-order approximation of the state deviation dynamics

$$H(t) = \delta u(t)^T \delta u(t) + \lambda^T(t) (A(t)\delta x(t) + B(t)\delta u(t)), \quad (17)$$

then the necessary conditions for optimality yields, for every time t ,

$$\dot{\lambda} = -\left(\frac{\partial H}{\partial \delta x}\right)^T = -A^T \lambda \quad (18)$$

$$0 = \frac{\partial H}{\partial \delta u} = 2\delta u + B^T \lambda \quad (19)$$

$$\delta \dot{x} = \left(\frac{\partial H}{\partial \lambda}\right)^T = A\delta x + B\delta u \quad (20)$$

The optimal $\delta u(t)$ can be obtained through forward integrating $\Phi(t, t_0)$ to $t = t_f$ by

$$\dot{\Phi}(t, t_0) = \begin{bmatrix} A(t) & -0.5B(t)B^T(t) \\ 0 & -A^T(t) \end{bmatrix} \Phi(t, t_0), \text{ subject to } \Phi(t_0, t_0) = I \quad (21)$$

$$\lambda(t_0) = \Phi(t_f, t_0)_{12}^{-1} \left(-\Phi(t_f, t_0)_{11} \delta x(t_0) \right) \quad (22)$$

$$\lambda(t) = \Phi(t_f, t_0)_{21} \delta x(t_0) + \Phi(t_f, t_0)_{22} \lambda(t_0) \quad (23)$$

$$\delta u(t) = -0.5B^T(t)\lambda(t). \quad (24)$$

Again, this optimization problem is solved repeatedly for every time t , and only the first control is used. We note that this problem converges, at the limit, to the time-varying LQR with $Q = 0$, $R = I$, and $Q_f = \infty I$.

4 Results of Adding Feedback Control

We compare the results of the time-invariant LQR, time-varying LQR, and NOC in terms of following a same nominal trajectory when the process noise is present. The simulation is terminated at time t if the state deviates too far away from the nominal state ($\|\delta x(t)\|_2^2 > 1.0$). Tab. 1 reports the configuration of this experiment. Fig. 2 compares the deviation magnitude between these three feedback controllers. Fig. 3 show the position, control, and deviation trajectories for each feedback controllers. In summary,

Table 1: The experiment configuration. x_i is the normalized initial state, σ is the magnitude of the process noise, and Δt_c is the normalized time step for control update (equal to 6004.83 seconds).

g_0 [km/s ²]	C_1	C_2	C_3	l_{norm} [km]	t_{norm} [sec]	a_{norm} [km/s ²]	x_i	σ	Δt_c
3.1244e-11	2.0	0.0	0.0	52.4409	6.00483e+6	1.45438e-12	(0.0019, 0.0488, 0.0, 0.0, 0.0, 4.7163)	$\text{diag}([0.0, 0.0, 0.0, 0.05, 0.2, 0.2])$	0.001 (=6004.83 sec)

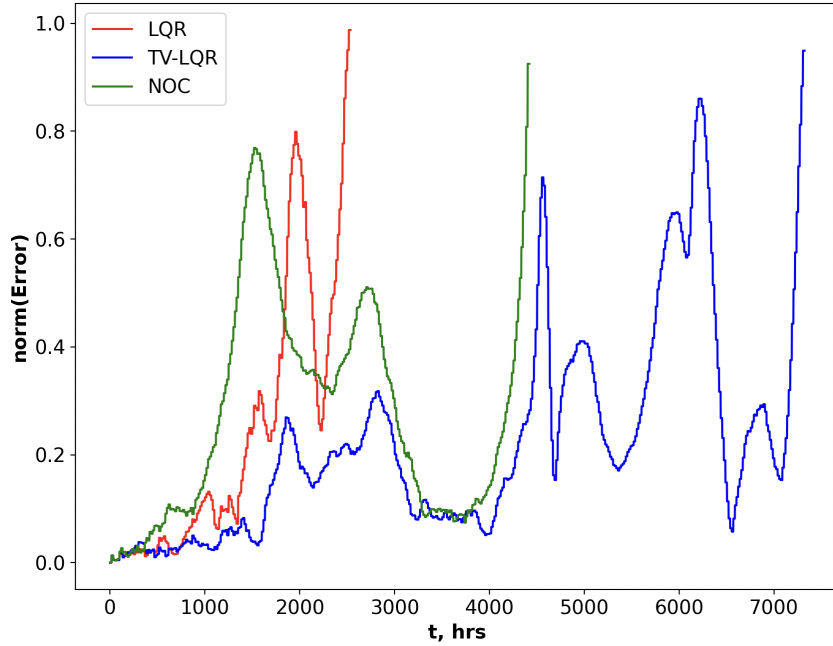


Figure 2: The magnitude of the state deviation over time for the time-invariant LQR (the red line), the time-varying LQR (the blue line), and the NOC (the green line).

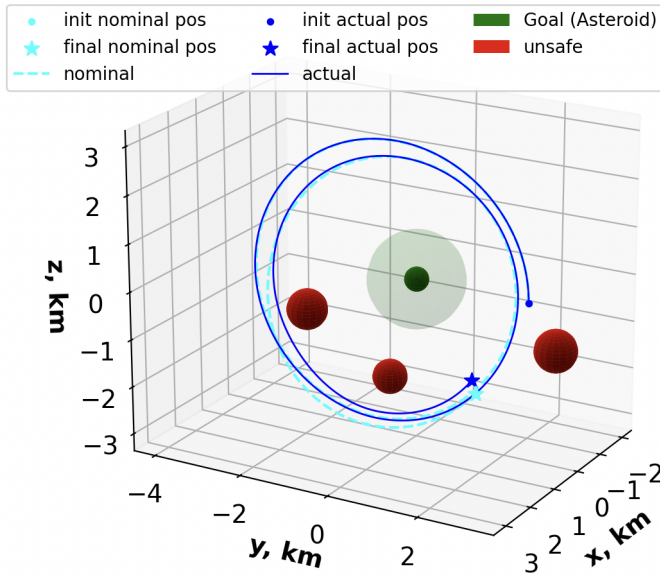
the time-varying LQR (the blue line) has the best result, while the time-invariant LQR (the red line) has the worst performance as expected.

From this experiment, we observe that the state will eventually deviate from the nominal trajectory. This is expected due to two reasons (i) all the discussed feedback controllers approximate the state deviation dynamics as (time-invariant/time-varying) linear system though the true dynamics is nonlinear, and (ii) the optimization problems associated with these controllers do not explicitly take the process noise into account. It is likely to reduce the state deviation over time by (i) fine tuning the costs Q , R , Q_f or (ii) reducing the time step for control update Δt_c . However, this is not the focus of this work.

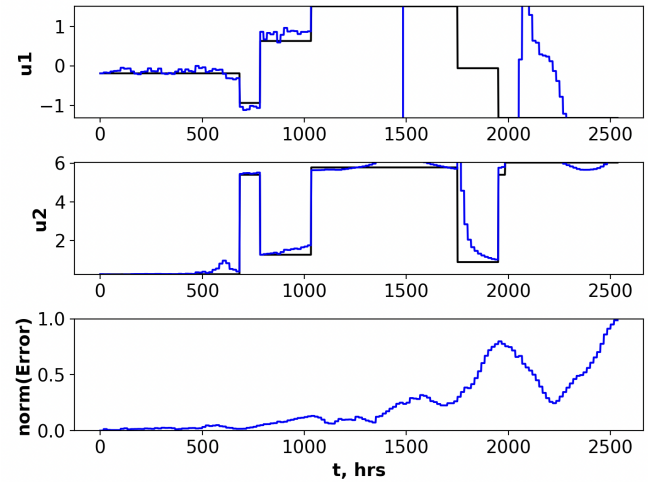
Recall that we want to control realistic systems (possibly modeled by Eq.1) in order to satisfy the *reach-avoid* specifications. Hence, any "sufficiently large" deviation would cause the system to either (i) not reaching the goals or (ii) hitting the unsafe regions. In fact, all of the feedback controllers tested above cannot drive the system to land on the asteroid when the state is simulated to the final time (i.e., without early termination). Following these observations, we present a simple and efficient approach to increase the chance of satisfying the *reach-avoid* specifications for general nonlinear systems subject to stochastic noise.

5 Control Policy with Re-planning

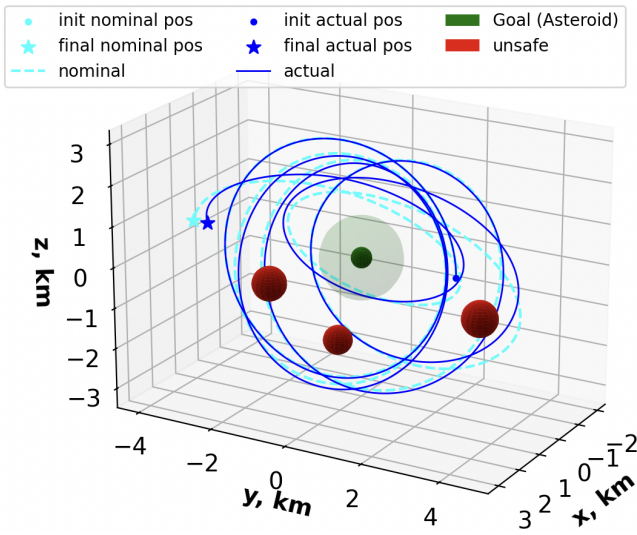
The idea is simple: for every time step, the feedback control will perform poorly if the state deviation is large. Hence, if the magnitude of the state deviation exceeds certain threshold at some time t , we re-plan a nominal trajectory that is able to safely reach the goals from the current state $x(t)$. Such re-planning can be achieved efficiently through sampling-based methods, enabling real-time applications.



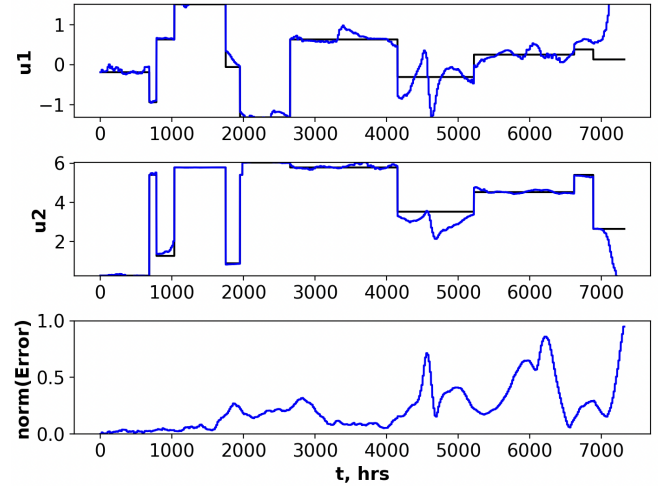
(a) LQR position



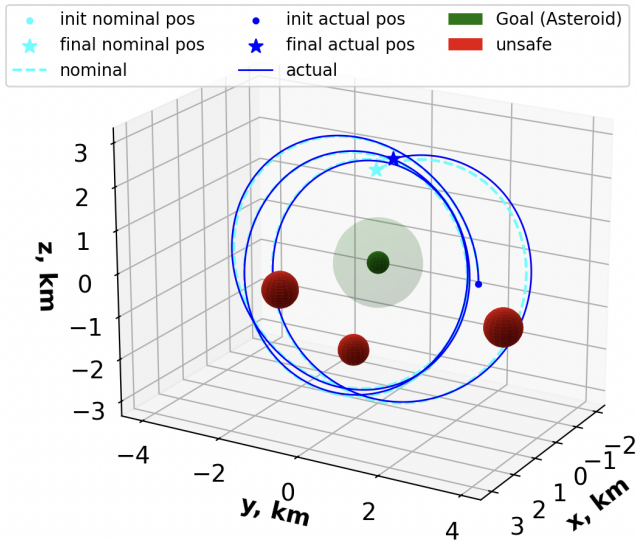
(b) LQR control and deviation



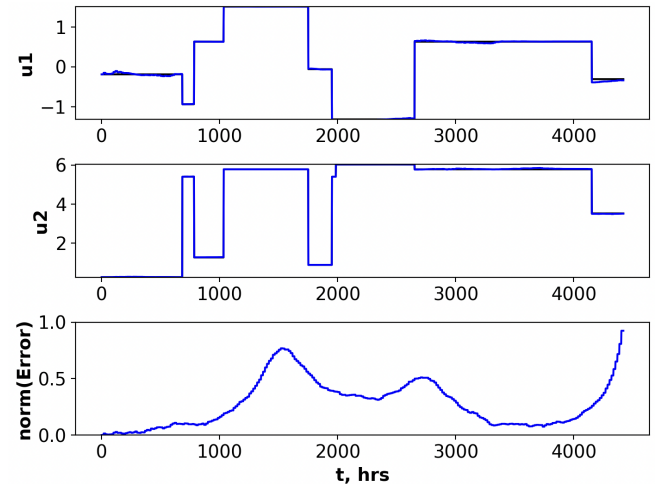
(c) TV-LQR position



(d) TV-LQR control and deviation



(e) NOC position



(f) NOC control and deviation

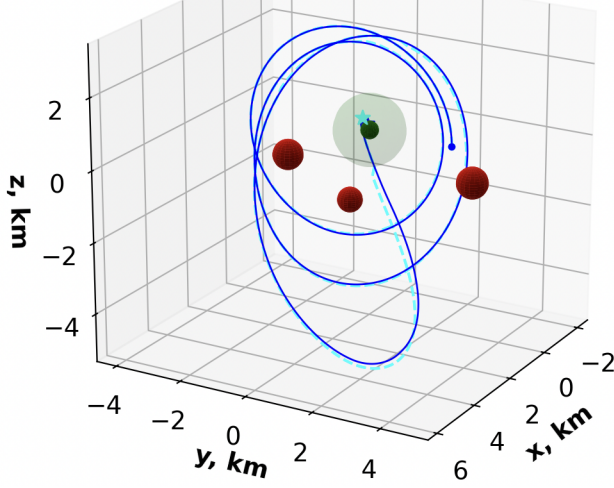
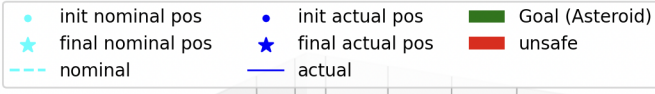
Figure 3: Position, control, and deviation trajectories for the time-invariant LQR, the time-varying LQR, and the NOC.

For the solar-sail landing problem, the feedback control is updated every 6004.83 seconds. The re-planning is triggered when $\|\delta x(t)\|_2^2 > 0.1$. The re-planning procedure has a maximum 60 seconds to find a initial nominal trajectory, and another 60 seconds (at maximum) to obtain better trajectory (by better, we mean smaller time of flight). This effectively introduces a input delay of 2% when the re-planning is executed. The following experiments are done by neglecting the input delay for simplicity. We note that this input delay could also be incorporated in the re-planning procedure to reduce its effect.

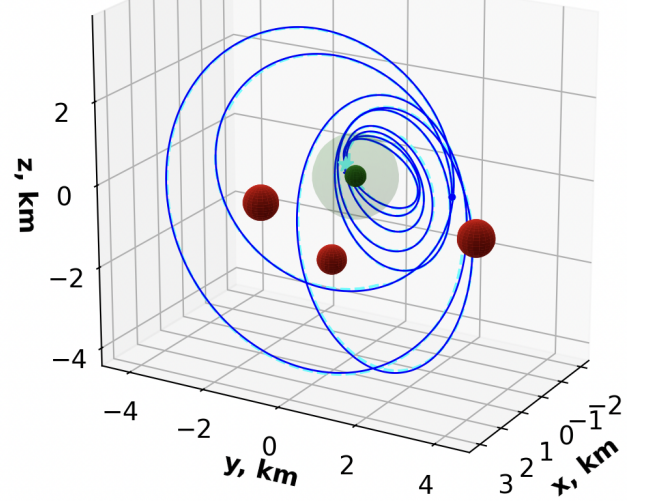
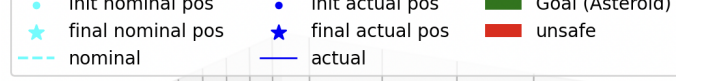
We first show a illustrative example using time-varying LQR and NOC as feedback controllers with SetRRT re-planning. Fig. 4 plots the position, control, and deviation trajectories. For both the TV-LQR and NOC, the state trajectory (blue solid line) follows the nominal trajectory (cyan dashed line), and eventually land safely on the asteroid. Although not visible, the nominal trajectory is composed of segments that are generated online at different time. In contrast, the nominal trajectory, without re-planning, consists of only one segment generated offline. Additionally, we observe that the TV-LQR control appears to be more stable when the state is close to the asteroid. Lastly, we see the periodically growth and resetting to zero in the deviation history, which is due to the re-planning. Note that all the $\|\delta x(t)\|_2^2$ is smaller than the user-specified threshold 0.1.

Secondly, Fig. 5 demonstrates successful landing when the magnitude of noise is increased up to 10 times. Due to the increase in uncertainty, the re-planning is triggered more frequently. Lastly, Fig. 6 plots the histogram of the final radius $\|x(t_f)\|_2$ in km (with red dotted line illustrating the radius of the asteroid surface) and the time of flight in hours. The experiment achieved 98% in satisfying the reach-avoid specifications. Both the failure cases are due to time-out of the re-planning process (recall that we set a maximum 60 seconds to find a initial trajectory), which leads to no update for the nominal trajectory (the state deviation cannot be reduced) and ultimately early termination $\|\delta x(t)\|_2^2 > 1.0$.

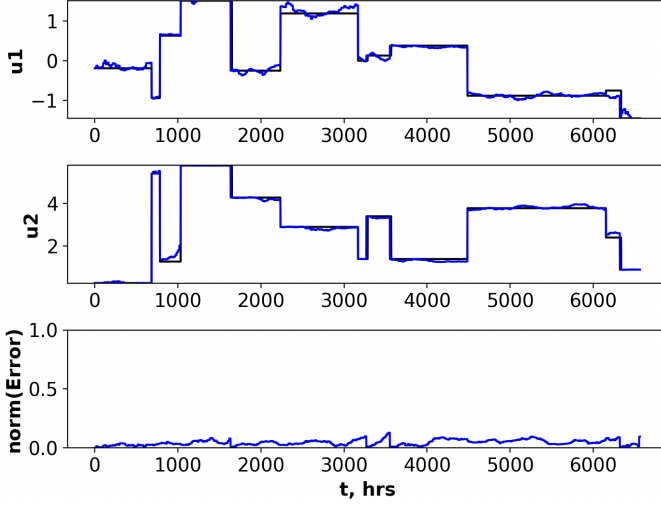
We note that either improving the feedback control or the re-planning computation will help boosting the empirical performance. The former reduces the frequency of triggering re-planning, and the latter increases the chance of finding a nominal trajectory given the planning time limit.



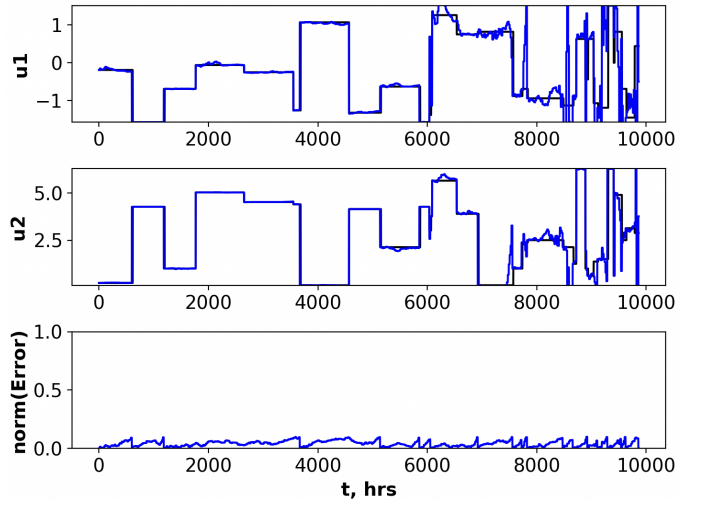
(a) TV-LQR trajectory



(b) NOC trajectory

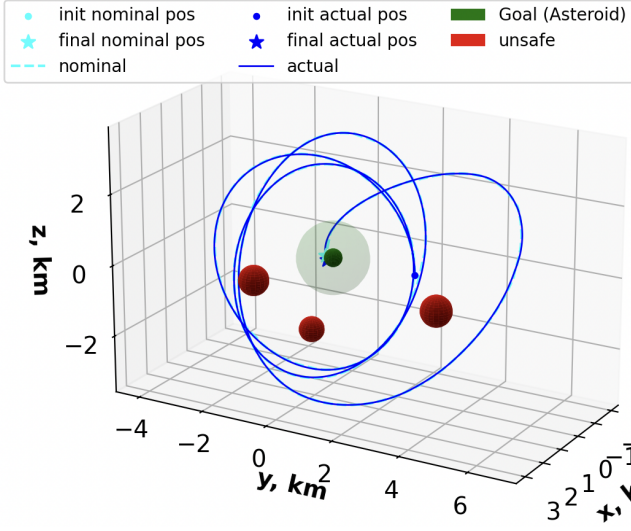


(c) TV-LQR control and deviation

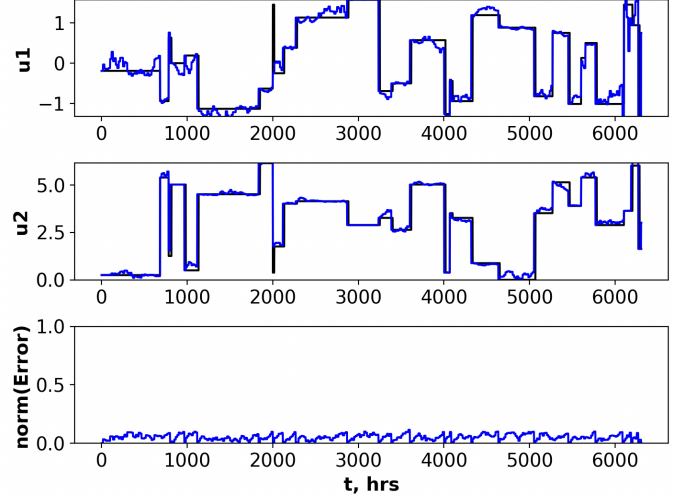


(d) NOC control and deviation

Figure 4: Position, control and deviation trajectories using TV-LQR/NOC and SetRRT re-planning.

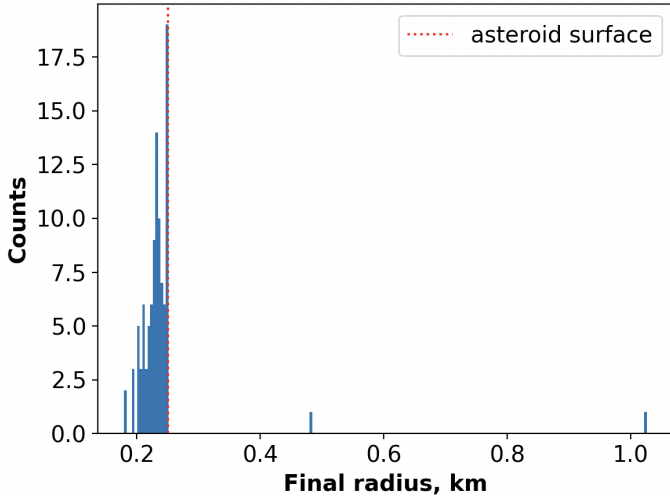


(a) Position trajectory

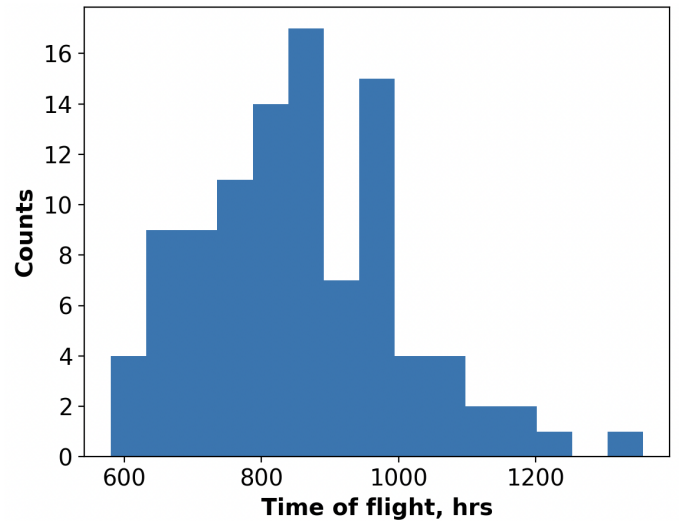


(b) Control and deviation trajectory

Figure 5: Results of the control policy (TV-LQR and SetRRT Re-planning) with larger process noise $\sigma = \text{diag}([0.0, 0.0, 0.0, 0.5, 0.5, 0.5])$.



(a) Final radius



(b) Time of flight

Figure 6: Histograms of 100 trials Monte Carlo using TV-LQR and SetRRT re-planning. From the left: 98/100 trials success to land safely.

6 Discussions

Although the proposed control policy demonstrate good empirical performance, we would like to discuss the guarantees and further research topics. Assuming that the dynamics is known, then the guarantees worth discussing is: **given the magnitude or characteristics of the stochastic noise, what is the probability of satisfying the specifications?**

We cannot answer this question with the current control policy because it does not *explicitly* account for uncertainty. Instead, it addresses the uncertainty *implicitly* by reducing the deviation, which is induced partially by the uncertainty and partially by linear dynamics approximations.

As a result, it is clear that the current control policy can be further improved by (i) incorporate more accurate dynamics and stochastic noise for the feedback control and (ii) consider stochastic noise for the nominal trajectory planner, which may lead to reasoning formal guarantees. Below, we discuss concrete examples.

6.1 Discussion on Further Research Topics

Consider nonlinear dynamics for feedback control

In the NOC, we use linear approximation for the deviation dynamics. Suppose we use the original nonlinear dynamics, then the Hamiltonian becomes $H(t) = \delta u^T(t)\delta u(t) + \lambda^T(t)\delta \dot{x}(t)$. By definition, $\delta x := x - x^*$, thus $\delta \dot{x} = f(x, u) - f(x^*, u^*)$. Hence, we could expand $f(x, u)$ at x^* and u^* to obtain a higher order approximation of the $\delta \dot{x}$. In addition, there exists theorems about the truncation errors and the region of convergence.

Consider stochastic noise for feedback control

Covariance control [4] can be used to take the stochastic noise into account. However, it is worth noted that most covariance control methods consider (time-varying) linear dynamics. It would be non-trivial to reason formal guarantee for nonlinear dynamics.

Incorporate stochastic noise for feed-forward planner

We could inform the SetRRT planner with stochastic noise and the feedback controller. For instance, the procedure of adding a node may become:

- For a node x , sample u^* and time duration to obtain a new state x' .
- Sample N perturbed states from the region $B_r(x)$. Use the feed-forward u^* and the feedback controller δu , the same time duration, and random noise w to simulate N samples of the perturbed new states.
- If p_1 percent of these N new states are within the ball $B_r(x')$, then add a new node with state x' . Here, more complicated adding conditions could be used, such as control constraints on $u^* + \delta u$.
- Keep doing it until a state arrives goal regions (or more complicatedly, the corresponding N perturbed new states are d -close to the goal regions with p_2 probability).

By doing so, we may be able to reason the probability of re-planning, as well as specification satisfaction.

6.2 Discussion on Application Value

We argue that the presented control policy has good application value. For a given application, one could start from the most simple feedback controller and re-planner, and conduct Monte-Carlo simulations. If the results is satisfactory (k out of N trials success), then great. If not satisfied, one could improve

each components with more complicated formulation and redo the Monte-Carlo. Through this process, one is likely to obtain a control policy that is "just complex" enough to satisfy the specifications.

References

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